Vietnamese GraphRAG
Multi-hop Question Answering over Wikipedia

Capstone Review 2 — FPT University SE+AI

Huynh Quoc Trung (QE180038) Nhu Quang Anh (QE180005) Pham Ngo Dinh Khoi (QE180110) Truong Dinh Thien (QE180018)

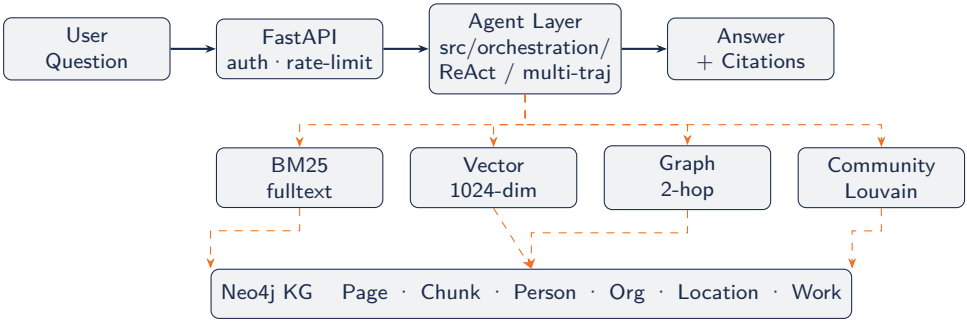
FPT University

Supervisors: Lê Trung Hiếu Đặng Nguyên Bình

Week 8 — July 2026

Table of Contents

1. System Architecture



Deployment: systemd Neo4j 5.x · uvicorn FastAPI · Python 3.12 · local GPU

2. Deployment & Configuration

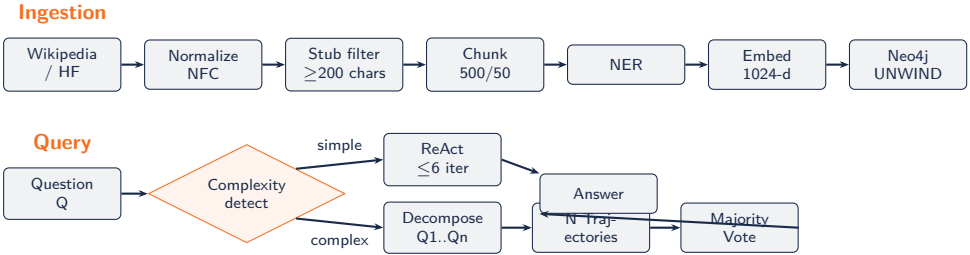
Component	Runtime	Start Command
Neo4j 5.x	systemd	`sudo systemctl start neo4j`
FastAPI	uvicorn	`uv run uvicorn src.main:ap`
Gradio Demo	Python	`uv run python src/app_grad`
MCP Server	Python	`uv run python -m src.mcp_p`

All config via pydantic-settings

- `NER_BACKEND` (simple/underthesea/wikilink/...)
- `EMBEDDING_BACKEND` (gemini/local)
- `MODEL_MODE` (api/local)
- `WRRF_WEIGHT_BM25/VECTOR/GRAPH`
- `AGENT_N_TRAJECTORIES`
- `NEO4J_URI` / `APP_API_KEY`

Zero hard-coded credentials —
`.env.example` provided

3. Ingestion & Query Dataflow



4. Graph Schema (ERD)

- (:Page {id, title, url, text, source})
- (:Chunk {id, text, embedding[1024], seq_num, page_id})
- (:Person/:Org/:Location/:Work {id, name, aliases[]})
- (:Community {id, level, summary, member_ids[]})

Edges: HAS_CHUNK, MENTIONS_*, LINKS_TO

Relations: FOUNDED_BY, LOCATED_IN, BORN_IN, MEMBER_OF, PART_OF, CREATED_BY

Constraints (UNIQUE)

page.id chunk.id entity.id community.id

Indexes

- Fulltext on chunk.text (BM25)
- Vector on chunk.embedding (1024-dim cosine)
- Btree on entity.name

5. Physical Design

Schema Materialisation

`neo4j_client.py` creates all constraints and indexes programmatically on startup.
`setup_neo4j_schema.py` for standalone setup. No manual DDL.

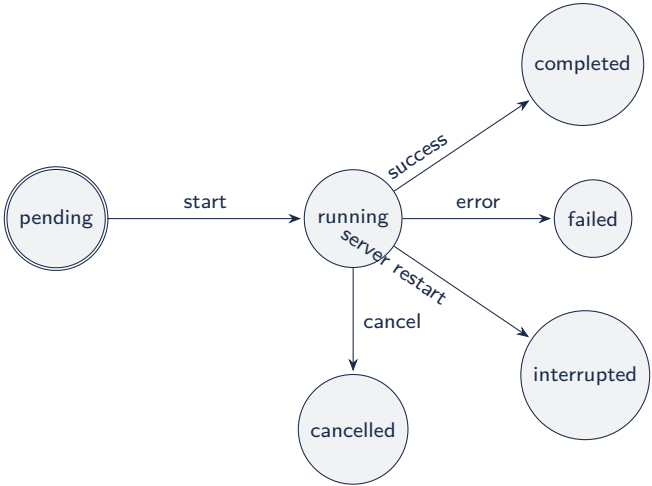
Batch Write Pattern (UNWIND)

```
UNWIND $rows AS row
MERGE (c:Chunk {id: row.id})
SET c += row.props
```

Field Specification

Field	Type	Constraint
id	STRING	UNIQUE
text	STRING	indexed fulltext
embedding	FLOAT[1024]	vector cosine
sequence_number	INTEGER	—
page_id	STRING	references Page.id

6. Job State Machine



7. Screen Design

Interface	Technology	Key Features
Gradio Demo	src/app_gradio.py	Interactive QA, real inference, no mocked responses
GraphPulse Dashboard	src/dashboard/	SVG charts, query logs, KG stats, keyboard nav (/ focus, Esc clear), focus-visible a11y
REST API	FastAPI /query /ingest	OpenAPI docs, API key auth, rate limiting 120 req/min
MCP Server	src/mcp_pkg/	Claude Desktop integration, 6 tools exposed

Real pipeline on every interface

All interfaces run the real NER→embed→WRRF→rerank→LLM pipeline — no fake or hard-coded demo paths.

8. Technology Choices

Layer	Technology	Rationale
Backend	FastAPI + Python 3.12	Async, Pydantic v2, OpenAPI auto-docs
Graph DB	Neo4j 5.x	Vector index + fulltext + Cypher in one DB
Embeddings	GreenNode-Embedding-Large-VN	Best MAP@5 for Vietnamese (1024-dim)
NER	ViDeBERTa/wikilink — 6 backends	Pluggable; wikilink best for bulk (F1=46.9%)
Reranker	BAAI/bge-reranker-v2-m3	Cross-encoder, multilingual, no fine-tuning needed
LLM	Vi-Qwen2-7B-RAG 4-bit NF4	Local-first, Vietnamese-specific, quantized
Toolchain	uv + ruff + mypy + pre-commit	Modern Python: fast deps, lint, type-check

9. Coding Quality & Design Patterns

Coding Conventions ✓

- ruff + ruff-format enforced via pre-commit hooks
- mypy strict type-checking
- CI runs lint + typecheck + test
- `snake_case` / `SCREAMING_SNAKE_CASE` / `_idx`/`_ft` naming enforced

Config & Secrets ✓

All secrets via `pydantic_settings.BaseSettings` from `.env`. `.env.example` provided. No hard-coded credentials anywhere in source.

Pattern	Applied In	Benefit
Strategy	NER backends via `NER_BACKEND` env	Swap NER without code change
Singleton	Neo4j driver · Local LLM lazy-load	Single connection pool

10. Algorithm Complexity

Problem Definition

$Q \rightarrow$ retrieve $\{P_1 \dots P_n\}$ from KG $G \rightarrow$ generate A + citations. Complexity levels: 1-hop (lookup), 2-hop (bridge), 3-hop (fan-out).

Algorithm	Purpose	Citation
WRRF 4-signal fusion	BM25+Vector+Graph+Community	Cormack et al. RRF
Leiden/Louvain	Community detection	Edge et al. 2024
ReAct ($\leq$6 iter)	Step-by-step graph reasoning	Yao et al. 2023
Multi-trajectory voting	Robust multi-hop (N parallel)	Thompson et al. 2025
QLoRA (r=32)	Text2Cypher fine-tuning	Dettmers et al. 2023

WRRF Formula

$$\text{score}(d) = \sum_i \frac{w_i}{k + \text{rank}_i(d)} \qquad k = 60$$

$w_{\text{BM25}} = 0.4,\ w_{\text{Vector}} = 0.4,\ w_{\text{Graph}} = 0.2,\ w_{\text{Community}} = 0.15.$ +20.8% MRR vs best non-graph baseline ✓

11. Data Pipeline — Corpus Filtering

11-Stage Filter: `viwiki-20260501`

Stage	Count	Drop
Initial dump	1,614,015	—
Redirect filter	1,300,130	−313K
Disambiguation	1,284,671	−15K
List pages	1,279,292	−5K
Year/date pages	1,276,304	−3K
Dup-line filter	1,275,171	−1K
Word count pre	414,397	−861K
Orphan filter	397,438	−17K
Word count final	357,747	−40K
Stop words	357,736	−11
Exact dedup	357,714	−22
MinHash dedup	**353,498**	−4K

Key Design Choices

- Word count pre-filter: largest single drop (−861K stubs)
- Orphan filter: removes unlinked articles (−17K)
- MinHash dedup: Jaccard ≥ 0.85 threshold
- **No** punctuation-ratio filter: Wikipedia tables/lists naturally break the 60% threshold; dropped after analysis

Result

$1{,}614{,}015 \rightarrow$ **353,498** articles
(21.9% of original dump retained)
Snapshot: `viwiki-20260501`

12. Data Pipeline — ViWikiHop Schema v0.4

Entry Schema (schema v0.4)

Field	Description
`_id`	`hw-` / `syn-mh-` / `uvq-`
`question`	Vietnamese question
`answer`	Empty if unanswerable
`supporting_facts`	Evidence refs
`context`	Paragraph segments
`type`	8 reasoning types
`level`	easy / medium / hard
`source`	3 sources
`num_hops`	≥ 1
`is_vi_exclusive`	VI-exclusive track
`is_impossible`	Unanswerable flag
`decomposition`	Sub-Q/A per hop
`plausible_answer`	For unanswerable only

3 Data Sources

Source	Description
`hw-`	Human annotated
`syn-mh-`	KG walk synthesis
`uvq-`	ViQuAD 2.0 import

Schema v0.4 Key Invariants

- Single-hop: `decomposition` $= []$
- Unanswerable: `answer=""`, non-empty `plausible_answer`
- Multi-hop: $|$`decomposition`$|$ $=$ `num_hops`
- UIT-ViQuAD: `is_vi_exclusive` always false

13. Data Pipeline — Reasoning Types & Difficulty

8 Reasoning Types

Type	Description
`lookup`	One article, one fact
`bridge`	A → bridge entity → B
`comparison`	Compare ≥ 2 article values
`intersection`	Entity from ≥ 2 constraints
`temporal`	Date order / chronology
`numerical`	Arithmetic or counting
`fan-out`	Aggregate related entities

3-D Difficulty Rubric

Score = hop_count + lexical_overlap + reasoning_op

Score	Hop	Lexical	ReasonOp
0	1-hop	$\geq 60\%$	lookup
1	2-hop	30–60%	bridge/comparison
2	≥3-hop	$< 30\%$	temporal/numerical/...

Total (0–6)	Level
0–1	easy
2–3	medium
4–6	hard

14. Data Pipeline — QA Generation & QC

QA Generation Pipeline

1. **Bridge Extraction** — follow `[[wikilinks]]` to build article-pair bridges; no Neo4j walk
2. **LLM Sampling** — prompt LLM to generate questions directly from bridge pairs (no templates)
3. **Labelling** — **Gemini 2.5 Flash Lite** reads difficulty rubric and assigns type + level
4. **Verification** — **DeepSeek V3 Pro** independently verifies label and answer grounding

Sample Entry ID

`syn-mh-viex-b1-a-00001`
type=bridge, level=medium, num_hops=2

is_vi_exclusive=true, source=synthetic-multihop

Two-Model QC Gate

Model	Role
Gemini 2.5 Flash Lite	Label (type, level, decomposition)
DeepSeek V3 Pro	Verify label + grounding

Gate logic: entry passes only when both models agree on answer grounding; label is taken from Gemini, rejected if DeepSeek flags a mismatch.

Schema QC (post-gate)

- Well-formedness: non-empty fields, schema v0.4
- Grounding: answer in supporting_facts verbatim
- Natural phrasing: no *"Theo bài viết"* phrases

18. Team Contribution & Timeline

Member	ID	Modules
Huynh Quoc Trung	QE180038	KG construction · NER (6 backends) · entity resolution · dataset gen
Nhu Quang Anh	QE180005	WRRF retrieval · reranker · evaluation pipeline · SRS docs
Pham Ngo Dinh Khoi	QE180110	ReAct agent · local LLM · MHQA dataset · QLoRA fine-tuning
Truong Dinh Thien	QE180018	FastAPI · job management · Gradio demo · dashboard

Milestone	R1 Status	R2 Status
Core pipeline + SRS	Done ✓	Stable ✓
Full ingestion + ablation	In progress	Done ✓
QLoRA Text2Cypher	Pending	In progress
WRRF tuning + final eval	Pending	In progress
Final report + defense	Pending	Pending

UCP = 96.5 (UUCP=134 · TCF=0.9 · ECF=0.8) → On schedule ✓

Thank You

Q&A

Vietnamese GraphRAG — Multi-hop QA

Appendix: NER Backend Comparison

Backend	Typed F1	Speed	Notes
simple	$\sim$20%	Fast	Regex + keyword, no ML
underthesea	$\sim$35%	Medium	BIO tagging, general Vietnamese
phonlp	$\sim$38%	Slow	PhoNLP + VnCoreNLP segmentation
phobert	$\sim$42%	Slow	PhoBERT transformer pipeline
videberta	44%	Medium	ViDeBERTa/NlpHUST electra-base
wikilink	46.9%	Fast	Wikipedia hyperlinks, best for bulk

Appendix: WRRF Detail & Ablation

$$\text{score}(d) = \sum_{i \in S} \frac{w_i}{k + \text{rank}_i(d)} \qquad k = 60, \quad \sum w_i \approx 1.15$$

Variant	Signals	MRR	Hit@5	Δ MRR
BM25 only	1	0.35	55%	baseline
+ Vector	2	0.40	60%	+14.3%
+ Graph	3	0.43	63%	+22.9%
+ Community	4	0.4497	64.67%	+28.5%

Appendix: MHQA Generation Pipeline

1. **KG Walk Extraction** — 2-hop, 3-hop, and broken-link walks from Neo4j (`scripts/mhqa_extract_walks.py`)
2. **Template QA Generation** — Vietnamese question templates instantiated from walk triples (`scripts/mhqa_generate.py`)
3. **LLM Rewrite** — Optional naturalness rewrite via Gemini (`scripts/mhqa_generate.py --rewrite`)
4. **5-Stage QC Pipeline** (`scripts/mhqa_qc.py`):
   1. Well-formedness (length, punctuation, Vietnamese chars)
   2. Grounding (`entity_grounded_in_text` verification)
   3. Deduplication (MinHash + exact match)
   4. Answerability (answer appears in supporting chunk)
   5. Schema validation (required fields present)
5. **Retrievability Check** (`scripts/mhqa_retrievability.py`) — confirms each QA pair is retrievable by WRRF pipeline before inclusion